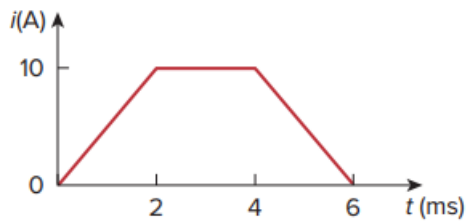


## Problem

The current through a 5-mH inductor is shown in Fig. 6.66. Determine the voltage across the inductor at  $t = 1$ , 3, and 5 ms.

**Fig. 6.66:**



## Step-by-step solution

## Step 1 of 5

Refer to Figure 6.66 in the textbook.

Write the line equation.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

Write the line equation for  $0 \leq t < 2 \text{ ms}$ .

$$\frac{10 - 0}{2 \times 10^{-3} - 0} = \frac{i(t) - 0}{t - 0}$$

$$i(t) = 5 \times 10^3 t$$

Write the line equation for  $2 \text{ ms} \leq t < 4 \text{ ms}$ .

$$i(t) = 10$$

Write the line equation for  $4 \text{ ms} \leq t < 6 \text{ ms}$ .

$$\frac{0 - 10}{6 \times 10^{-3} - 4 \times 10^{-3}} = \frac{i(t) - 10}{t - 4 \times 10^{-3}}$$

$$-5 \times 10^3 = \frac{i(t) - 10}{t - 4 \times 10^{-3}}$$

$$i(t) - 10 = -5 \times 10^3 t + 20$$

$$i(t) = -5 \times 10^3 t + 30$$

## Step 2 of 5

Write the current  $i(t)$  expression for different time intervals.

$$i(t) = \begin{cases} 5 \times 10^3 t & 0 \leq t < 2 \text{ ms} \\ 10 & 2 \text{ ms} \leq t < 4 \text{ ms} \\ 30 - 5 \times 10^3 t & 4 \text{ ms} \leq t < 6 \text{ ms} \end{cases}$$

Consider  $L = 5 \text{ mH}$ .

Consider the equation for voltage across the inductor.

$$v(t) = L \frac{di}{dt} \dots\dots (1)$$

**Step 3** of 5

Determine the voltage across inductor at  $t = 1 \text{ ms}$ .

Replace  $L = 5 \text{ mH}$  and  $i(t) = 5 \times 10^3 t$  in equation (1).

$$\begin{aligned} v(t) &= L \frac{di}{dt} \\ &= (5 \times 10^{-3}) \frac{d}{dt} (5 \times 10^3 t) \\ &= (5 \times 10^{-3}) (5000) \\ &= 25 \text{ V} \end{aligned}$$

Therefore, the voltage at  $t = 1 \text{ ms}$  is 25 V.

**Step 4** of 5

Determine the voltage across inductor at  $t = 3 \text{ ms}$ .

Replace  $L = 5 \text{ mH}$  and  $i(t) = 10$  in equation (1).

$$\begin{aligned} v(t) &= L \frac{di}{dt} \\ &= (5 \times 10^{-3}) \frac{d}{dt} (10) \\ &= (5 \times 10^{-3}) (0) \\ &= 0 \text{ V} \end{aligned}$$

Therefore, the voltage at  $t = 3 \text{ ms}$  is 0 V.

**Step 5** of 5

Determine the voltage across inductor at  $t = 5 \text{ ms}$ .

Replace  $L = 5 \text{ mH}$  and  $i(t) = 30 - 5 \times 10^3 t$  in equation (1).

$$\begin{aligned} v(t) &= L \frac{di}{dt} \\ &= (5 \times 10^{-3}) \frac{d}{dt} (30 - 5 \times 10^3 t) \\ &= (5 \times 10^{-3}) (-5 \times 10^3) \\ &= -25 \text{ V} \end{aligned}$$

Therefore, the voltage at  $t = 5 \text{ ms}$  is -25 V.